Now look at the code in `CH7-09-Production Batch Inference`. The big changes include the `scoring_df`, the dataframe we apply our model to for predictions. Notice in *Figure 7.23* that the `min_time` and `max_time` variables provide boundaries for the transactions, ensuring the batch feature values have been calculated. Additionally, the inference table also provides a boundary preventing duplicate prediction calculations.

```
min_time = sql(f"SELECT MIN(LookupTimestamp) FROM {ft_name}").collect()[0][0]
max_time = sql(f"SELECT MAX(LookupTimestamp) FROM {ft_name}").collect()[0][0]

if not spark.catalog.tableExists(inference_table) or spark.table(tableName=inference_table).isEmpty
():
  scoring_df = sql(f"""
                   SELECT Amount,CustomerID,Product,TransactionTimestamp FROM {table_name}
                   WHERE TransactionTimestamp <= '{max_time}' AND TransactionTimestamp >= '
                   {min_time}'
                   """)
  sql(f"""
      CREATE TABLE IF NOT EXISTS {inference_table}
      (Amount FLOAT, CustomerID INT, Product STRING, TransactionTimestamp TIMESTAMP,
      transactionCount INT, isTimeout BOOLEAN, MaxProductAmount FLOAT, MaxDifferenceRatio FLOAT,
      prediction DOUBLE, model_version STRING, actual_label INT) USING delta TBLPROPERTIES (delta.
      enableChangeDataFeed = true)
  """)
else:
  last_inf_time = sql(f"SELECT MAX(TransactionTimestamp) FROM {inference_table}").collect()[0][0]
  scoring_df = sql(f"""
                   SELECT Amount,CustomerID,Product,TransactionTimestamp FROM {table_name}
                   WHERE TransactionTimestamp <= '{max_time}' AND TransactionTimestamp >= '
                   {min_time}'
                   AND TransactionTimestamp > '{last_inf_time}'
                   """)
```

Figure 7.23 – The scoring_df query configuration

The inference table in `CH7-08` needs to be updated to fit the requirements for the inference table monitoring provided by Lakehouse Monitoring. This means adding the `model_version` and `actual_label` columns. The `actual_label` column is set to NULL so that it is clear the value has not been updated yet; see *Figure 7.24*.

```
scored = fe.score_batch(
  model_uri=f"models:/{model_name}/{model_version}",
  df=scoring_df
)
scored.withColumn("model_version",lit(model_version))\
  .withColumn(colName="actual_label",col=lit(None))\
  .write.mode('append').format('delta').saveAsTable(inference_table)
```

Figure 7.24 – The addition of the model version and actual label columns